# Power Sums, a Recurrence Basis, and Substitution Identities

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# Introduction

This note develops a self-contained algebraic framework for power sums $s_n = x^n + y^n + z^n$ and their relatives. The central object is a sequence $K(n)$ built from the antisymmetric weights of $x, y, z$; together with two companion sequences $L$ and $M$, it forms a basis for all sequences satisfying the same linear recurrence as the power sums.

All arguments use only the recurrence and elementary algebra; no advanced number theory or analysis is required.

# 1 Setup

Throughout, $x$, $y$, $z$ are distinct and nonzero complex (or real) numbers. We write their elementary symmetric polynomials as

$$e_1 = x + y + z, \qquad e_2 = xy + yz + zx, \qquad e_3 = xyz,$$

and define the *minimal polynomial*

$$P(t) = t^3 - e_1 t^2 + e_2 t - e_3 = (t - x)(t - y)(t - z).$$

Since $e_3 = xyz \neq 0$ we may divide by any power of $x$, $y$, or $z$, and in particular extend all recurrences to all integers $n \in \mathbb{Z}$.

## Cyclic sum notation

Many expressions in this paper are cyclic in $x, y, z$. We write

$$[g(x)]_{\text{cyc}} \;:=\; g(x) + g(y) + g(z)$$

for the cyclic sum of any expression $g$, where the implicit substitution rule is $x \to y \to z \to x$. For example, $e_1 = [x]_{\text{cyc}}$, $e_2 = [xy]_{\text{cyc}}$, $e_3 = xyz$, and more usefully:

$$[x^n(y - z)]_{\text{cyc}} \;=\; x^n(y - z) + y^n(z - x) + z^n(x - y).$$

This notation eliminates the need to write all three terms of any cyclic identity.

# 2 Weighted Power Sums

**Definition 2.1.** For any integer $n$ and a weight triple $(k, l, m)$, define

$$f_n(k, l, m) = kx^n + ly^n + mz^n = [k\, x^n]_{\text{cyc}}\,,$$

where in the cyclic notation it is understood that the weights cycle as $(k, l, m) \to (l, m, k)$ alongside $(x, y, z) \to (y, z, x)$.

Special cases of immediate interest:

$$s_n \;:=\; f_n(1,1,1) \;=\; x^n + y^n + z^n \;=\; [x^n]_{\text{cyc}}\,.$$

The key properties of $f_n$ are linearity in weights and the universal recurrence inherited from $P(t)$.

**Theorem 2.2** (Recurrence)**.** *For any fixed weights $(k,l,m)$ and any integer $n$,*

$$f_{n+3}(k,l,m) = e_1\, f_{n+2}(k,l,m) - e_2\, f_{n+1}(k,l,m) + e_3\, f_n(k,l,m).$$

*Proof.* Each of $x, y, z$ satisfies $t^3 = e_1 t^2 - e_2 t + e_3$. Multiply by $t^n$, weight by $k, l, m$ respectively, and sum. □

Because $e_3 \neq 0$ the recurrence can also be run backwards, making $f_n(k,l,m)$ well-defined for all $n \in \mathbb{Z}$.

# 3 The Basis $K$, $L$, $M$

Any sequence satisfying the recurrence of Theorem 2.2 is determined by its values at three consecutive integers. We single out the three basis sequences with initial values forming the identity matrix at $n = 0, 1, 2$.

## 3.1 Definition via the antisymmetric weights

The triple of *difference weights*

$$(w_x,\, w_y,\, w_z) \;:=\; (y-z,\; z-x,\; x-y)$$

satisfies $w_x + w_y + w_z = 0$ and is nonzero since $x, y, z$ are distinct. A short computation gives

$$f_0(w_x, w_y, w_z) = 0, \quad f_1(w_x, w_y, w_z) = 0, \quad f_2(w_x, w_y, w_z) = \big[x^2(y-z)\big]_{\text{cyc}} =: \Delta,$$

where $\Delta = -(y-z)(z-x)(x-y) \neq 0$. (The $f_2$ identity follows by straightforward algebra, or see Section 6.)

**Definition 3.1.** For any integer $n$, define

$$K(n) \;=\; \frac{f_n(w_x,\, w_y,\, w_z)}{\Delta} \;=\; \frac{[x^n(y-z)]_{\text{cyc}}}{[x^2(y-z)]_{\text{cyc}}}.$$

**Definition 3.2.**

$$L(n) = K(n+1) - e_1\, K(n), \qquad M(n) = e_3\, K(n-1).$$

The initial values follow immediately:

| $n$ | 0 | 1 | 2 |
|---|---|---|---|
| $K(n)$ | 0 | 0 | 1 |
| $L(n)$ | 0 | 1 | 0 |
| $M(n)$ | 1 | 0 | 0 |

The triple $(K, L, M)$ forms the *standard basis* for all integer sequences satisfying the recurrence: the columns at $n = 0, 1, 2$ are exactly the identity matrix. All three sequences satisfy the same recurrence as $f_n$, since $K$ inherits it from $f_n$, and $L$, $M$ are linear combinations of shifts of $K$.

**Remark 3.3.** The explicit values of $K$ at small $n$ are $K(2) = 1$, $K(3) = e_1$, $K(4) = e_1^2 - e_2$, and so on; $L$ and $M$ are recovered from $K$ via the definitions above. The backward extension gives $K(-1) = 1/e_3$, consistent with $M(0) = e_3 K(-1) = 1$.

# 4 The Decomposition Theorem

The following is the central engine of everything that follows.

**Theorem 4.1** (Decomposition)**.** *For any weights $(k, l, m)$ and any integer $n$,*

$$f_n(k, l, m) \;=\; K(n)\, f_2(k, l, m) \;+\; L(n)\, f_1(k, l, m) \;+\; M(n)\, f_0(k, l, m).$$

*Proof.* Both sides satisfy the same recurrence. They agree at $n = 0, 1, 2$ by the initial values of $K$, $L$, $M$. Uniqueness of recurrence solutions gives equality for all $n$. □

**Corollary 4.2** (Shift invariance)**.** *For any weights $(k, l, m)$ and any integers $n$, $j$,*

$$f_{n+j}(k, l, m) \;=\; K(n)\, f_{j+2}(k, l, m) \;+\; L(n)\, f_{j+1}(k, l, m) \;+\; M(n)\, f_j(k, l, m).$$

*Proof.* Fix $j$. The shifted sequence $h_n = f_{n+j}(k, l, m)$ satisfies the same recurrence, with $h_0 = f_j$, $h_1 = f_{j+1}$, $h_2 = f_{j+2}$. Apply Theorem 4.1 to $h_n$. □

The Decomposition Theorem says: *to evaluate any weighted power sum at index $n$, it suffices to know it at three consecutive indices — and the coefficients are always $K(n)$, $L(n)$, $M(n)$, regardless of the weights.*

# 5 Applications and Key Lemmas

## 5.1 Newton's identity: $s_n = x^n + y^n + z^n$

Apply the decomposition with weights $(1, 1, 1)$, noting $f_0(1, 1, 1) = 3$, $f_1(1, 1, 1) = e_1$, $f_2(1, 1, 1) = e_1^2 - 2e_2$:

$$s_n \;=\; K(n)\,(e_1^2 - 2e_2) \;+\; L(n)\, e_1 \;+\; 3\, M(n).$$

In particular, $s_0 = 3$, $s_1 = e_1$, $s_2 = e_1^2 - 2e_2$ match the known Newton values, and equality for all $n$ follows. Expanding the definitions yields the key formula:

**Lemma 5.1.** $s_n = e_1K(n+1) - 2e_2K(n) + 3e_3K(n-1)$ *for all integers* $n$.

*Proof.* Apply the Decomposition Theorem with weights $(1,1,1)$, substituting $f_0(1,1,1) = 3$, $f_1(1,1,1) = e_1$, $f_2(1,1,1) = e_1^2 - 2e_2$ and the definitions $L(n) = K(n+1) - e_1K(n)$, $M(n) = e_3K(n-1)$:

$$s_n = K(n)(e_1^2-2e_2)+L(n)\,e_1+3M(n) = K(n)(e_1^2-2e_2)+(K(n+1)-e_1K(n))e_1+3e_3K(n-1).$$

Expanding and cancelling the $e_1^2K(n)$ terms gives the result. □

**Theorem 5.2.** $s_n = K(n+2) + L(n+1) + M(n)$ *for all integers* $n$.

*Proof.* By Lemma 5.1, $s_n = e_1K(n+1) - 2e_2K(n) + 3e_3K(n-1)$. Substituting $K(n+2) = e_1K(n+1) - e_2K(n) + e_3K(n-1)$, $L(n+1) = K(n+2) - e_1K(n+1)$, and $M(n) = e_3K(n-1)$, one verifies that $K(n+2) + L(n+1) + M(n) = e_1K(n+1) - 2e_2K(n) + 3e_3K(n-1) = s_n$. □

# 6 Generalised $K$ and Substitution Identities

## 6.1 The generalised function $K_n(a,b,c)$

The formula in Definition 3.1 makes sense for any distinct nonzero triple $(a,b,c)$.

**Definition 6.1.** For distinct nonzero $a,b,c$ and any integer $n$,

$$K_n(a,b,c) \;=\; \frac{[a^n(b-c)]_{\text{cyc}}}{[a^2(b-c)]_{\text{cyc}}} \;=\; \frac{a^n(b-c)+b^n(c-a)+c^n(a-b)}{-(b-c)(c-a)(a-b)}.$$

So $K_n(x,y,z) = K(n)$. The function $K_n(a,b,c)$ satisfies the recurrence with symmetric polynomials $\varepsilon_i$ of $(a,b,c)$ in place of $e_i$, and has initial values $K_0 = K_1 = 0$, $K_2 = 1$ for any admissible triple.

The power of this definition is that *evaluating $K_n$ at a specific triple reduces, via the Decomposition Theorem, to a clean identity in $K$, $L$, $M$.* The next three subsections illustrate this with the three most natural substitutions.

## 6.2 The product substitution: $(a,b,c) = (yz, zx, xy)$

The triple $(yz, zx, xy)$ has symmetric polynomials

$$\varepsilon_1 = e_2, \qquad \varepsilon_2 = e_1e_3, \qquad \varepsilon_3 = e_3^2.$$

To evaluate $K_n(yz, zx, xy)$, note that by definition

$$\Delta \cdot K_n(yz,zx,xy) \;=\; [(yz)^n(zx-xy)]_{\text{cyc}} \;=\; f_n\big(z^{n-1}-y^{n-1},\; x^{n-1}-z^{n-1},\; y^{n-1}-x^{n-1}\big)\,.$$

Applying the Decomposition Theorem with these weights:

- $f_0 = 0$ (the weights sum to zero),

- $f_1 = [x^{n-1}(y-z)]_{\text{cyc}} = K(n-1)\cdot\Delta$,
- $f_2 = [x^{n-1}(y^2-z^2)]_{\text{cyc}} = -L(n-1)\cdot\Delta$ (both sides satisfy the same recurrence and agree at $n = 2, 3, 4$).

Therefore

$$\Delta \cdot K_n(yz, zx, xy) = K(n)\cdot(-L(n-1)\Delta) + L(n)\cdot(K(n-1)\Delta),$$

and dividing by $\Delta$:

**Theorem 6.2** (Product-substitution identity)**.** *For all integers $n$,*

$$K_n(yz,\, zx,\, xy) \;=\; K(n)\,L(n-1) - L(n)\,K(n-1).$$

Substituting $L(k) = K(k+1) - e_1K(k)$ and expanding, the $e_1$ terms cancel:

**Corollary 6.3** (Cassini identity)**.**

$$K(n)^2 - K(n+1)\,K(n-1) \;=\; K_n(yz,\, zx,\, xy).$$

**Remark 6.4.** This mirrors the classical Cassini identity $F_n^2 - F_{n+1}F_{n-1} = (-1)^{n-1}$ for Fibonacci numbers, where the right-hand side is a constant (up to sign). Here the right-hand side is the richer object $K_n(yz, zx, xy)$, which satisfies its own third-order recurrence with coefficients $(e_2, e_1e_3, e_3^2)$. In particular, $K(n)^2 - K(n+1)K(n-1)$ is a polynomial in $e_2$, $e_1e_3$, $e_3^2$ only — $e_1$ never appears alone.

## 6.3 The squaring substitution: $(a, b, c) = (x^2, y^2, z^2)$

The triple $(x^2, y^2, z^2)$ has symmetric polynomials

$$\varepsilon_1 = e_1^2 - 2e_2, \qquad \varepsilon_2 = e_2^2 - 2e_1e_3, \qquad \varepsilon_3 = e_3^2.$$

**Theorem 6.5** (Squaring-substitution identity)**.** *For all integers $n$,*

$$(e_3 - e_1e_2)\cdot K_n(x^2,\, y^2,\, z^2) \;=\; L(2n).$$

*Proof.* This is the special case $k = 2$ of Theorem 7.1 in Section 7, proved there in full. We verify the prefactor: $\Delta_2/\Delta = (x+y)(y+z)(z+x) = -(e_3 - e_1e_2)$, and at $k = 2$ the right-hand side of Theorem 7.1 is $K(2n)\,K(3) - K(2n+1)\,K(2) = e_1K(2n) - K(2n+1) = -L(2n)$, since $K(2) = 1$, $K(3) = e_1$, and $L(2n) = K(2n+1) - e_1K(2n)$. Combining signs gives the stated identity. □

Note $e_3 - e_1e_2 = -(x+y)(y+z)(z+x)$, the product of pairwise sums.

# 7 General Power Substitution

For an integer $k \geq 1$, write the *$k$-th difference discriminant* as

$$\Delta_k \;=\; -(y^k - z^k)(z^k - x^k)(x^k - y^k),$$

consistently with the sign convention $\Delta = \Delta_1 = -(y-z)(z-x)(x-y)$. The ratio $\Delta_k/\Delta$ is a symmetric polynomial in $x, y, z$; for example $\Delta_2/\Delta = (x+y)(y+z)(z+x) = -(e_3 - e_1 e_2)$.

**Theorem 7.1** (General power substitution)**.** *For all integers $k \geq 1$ and $n$,*

$$\frac{\Delta_k}{\Delta} K_n(x^k, y^k, z^k) \;=\; K(kn)\, K(k+1) - K(kn+1)\, K(k).$$

*Proof.* By definition, the numerator of $K_n(x^k, y^k, z^k)$ is

$$\left[(x^k)^n (y^k - z^k)\right]_{\text{cyc}} \;=\; f_{kn}(y^k - z^k,\, z^k - x^k,\, x^k - y^k),$$

since $(x^k)^n \cdot (y^k - z^k)$ is the $x$-term of $f_{kn}$ with weights $(y^k - z^k, z^k - x^k, x^k - y^k)$. Denote these weights $(p, q, r)$. We evaluate $f_0, f_1, f_2$ at $(p, q, r)$:

- $f_0(p, q, r) = 0$, since $p + q + r = 0$.
- $f_1(p, q, r) \;=\; \left[x(y^k - z^k)\right]_{\text{cyc}}$. Expanding the cyclic sum and collecting the $x^k$ terms (each $x \cdot y^k$ pairs with $-y \cdot x^k$ from the next term): $\left[x(y^k - z^k)\right]_{\text{cyc}} = \left[x^k(z - y)\right]_{\text{cyc}} = -\left[x^k(y - z)\right]_{\text{cyc}} = -K(k)\,\Delta$.
- $f_2(p, q, r) = \left[x^2(y^k - z^k)\right]_{\text{cyc}} = -\left[x^k(y^2 - z^2)\right]_{\text{cyc}}$. Writing $y^2 - z^2 = (y-z)(y+z) = -w_x(y+z)$ and expanding, $\left[x^k(y^2 - z^2)\right]_{\text{cyc}} = e_1\, f_k(w_x, w_y, w_z) - f_{k+1}(w_x, w_y, w_z) = (e_1 K(k) - K(k+1))\Delta = -L(k)\,\Delta$, so $f_2(p, q, r) = L(k)\,\Delta$.

Applying the Decomposition Theorem (with $kn$ in place of $n$):

$$f_{kn}(p, q, r) \;=\; K(kn) \cdot L(k)\,\Delta \;+\; L(kn) \cdot (-K(k))\,\Delta,$$

and since $f_{kn}(p, q, r) = K_n(x^k, y^k, z^k) \cdot \Delta_k$, dividing by $\Delta$:

$$\frac{\Delta_k}{\Delta} K_n(x^k, y^k, z^k) = K(kn)\, L(k) - L(kn)\, K(k).$$

Substituting $L(m) = K(m+1) - e_1 K(m)$ into both $L(k)$ and $L(kn)$ and expanding, the $e_1$ terms cancel, leaving $K(kn)\, K(k+1) - K(kn+1)\, K(k)$. □

## Special case $n = 1$

Since $K_1(a, b, c) = 0$ for any triple, the left-hand side vanishes, and the theorem gives $K(k)\, K(k+1) - K(k+1)\, K(k) = 0$: trivially true, and a consistency check.

### Special case $n = 2$

Since $K_2(a, b, c) = 1$ for any triple, the theorem gives the closed formula

$$\frac{\Delta_k}{\Delta} \;=\; K(2k)\,K(k+1) - K(2k+1)\,K(k).$$

### Special case $n = 3$

For any triple $(a, b, c)$ one has $K_3(a, b, c) = a + b + c$ directly from the closed form: the numerator $[a^3(b-c)]_{\text{cyc}}$ factors as $(a+b+c) \cdot [a^2(b-c)]_{\text{cyc}}$. Applied to $(x^k, y^k, z^k)$ this gives $K_3(x^k, y^k, z^k) = s_k$, and the theorem yields

$$\frac{\Delta_k}{\Delta}\, s_k \;=\; K(3k)\,K(k+1) - K(3k+1)\,K(k).$$

### Recovery of the squaring identity (Theorem 6.5)

Set $k = 2$. Then $K(2) = 1$ and $L(2) = K(3) - e_1K(2) = e_1 - e_1 = 0$, so $K(kn)\,L(k) - L(kn)\,K(k) = -L(2n)$. Since $\Delta_2/\Delta = (x+y)(y+z)(z+x) = -(e_3 - e_1e_2)$, Theorem 7.1 gives

$$-(e_3 - e_1e_2) \cdot K_n(x^2, y^2, z^2) \;=\; -L(2n),$$

which is exactly Theorem 6.5.

## 8 An Addition Formula for $K$

The following identity is used in Section 10 to relate $K(3n)$ and $K(3n+1)$ to $K(n)$ and $K(n+1)$.

**Lemma 8.1** (Addition formula)**.** *For all integers $m$ and $n$,*

$$K(m+n) \;=\; K(m)\,K(n+1) - e_3\,K(m-1)\,K(n-1) + \big(K(m+1) - e_1K(m)\big)\,K(n).$$

*Equivalently, using $L(m) = K(m+1) - e_1K(m)$ and $M(m) = e_3K(m-1)$,*

$$K(m+n) \;=\; K(m)\,K(n+1) + L(m)\,K(n) - M(m)\,K(n-1).$$

*Proof.* Fix $m$ and consider both sides as functions of $n$. Both satisfy the standard recurrence (in $n$) with coefficients $e_1, e_2, e_3$. At $n = 0, 1, 2$ the right-hand side gives, respectively,

$$K(m+0) = K(m), \quad K(m+1), \quad K(m+2),$$

matching the left-hand side. Uniqueness of recurrence solutions (Theorem 4.1) gives equality for all $n$. □

## 9 Consequences of $s_n = 0$ over the reals

Throughout this section $x$, $y$, $z$ are distinct nonzero real (or complex) numbers satisfying $s_n = x^n + y^n + z^n = 0$ for a fixed integer $n \geq 3$. No integrality is assumed.

## The ratio identity

Setting $k = n$ and $m = 3$ in Theorem 7.1 and using $K_3(x^n, y^n, z^n) = s_n = 0$ gives immediately:

**Corollary 9.1.** *If $s_n = 0$ then $K(3n)\,K(n+1) = K(3n+1)\,K(n)$.*

In other words the two ratios $K(3n)/K(n)$ and $K(3n+1)/K(n+1)$ are equal (whenever the denominators are nonzero). We write their common value $A = K(3n)/K(n)$.

## The common ratio $A$

**Theorem 9.2.** *If $s_n = 0$ and $K(n) \neq 0$, define $A = K(3n)/K(n)$. Then*

$$K(3n+1) = A\,K(n+1).$$

*Proof.* The ratio identity (Corollary 9.1) gives $K(3n)\,K(n+1) = K(3n+1)\,K(n)$. Substituting $K(3n) = A\,K(n)$ and dividing both sides by $K(n)$ yields the result. □

## The master identity

The two equalities above are the $j = 0$ and $j = 1$ cases of a uniform identity covering every offset $j$.

**Theorem 9.3** (Master identity)**.** *If $s_n = 0$ and $K(n) \neq 0$, then for all integers $j$,*

$$K(3n+j) \;=\; A\,K(n+j) \;+\; e_3^n\,K(j),$$

*where $A = K(3n)/K(n)$.*

*Proof.* Fix $n$. The right-hand side, viewed as a function of $j$, is a sum of two sequences: $j \mapsto A\,K(n+j)$ (a scalar multiple of a shift of $K$) and $j \mapsto e_3^n\,K(j)$ (a scalar multiple of $K$ itself). Both satisfy the recurrence with coefficients $e_1, e_2, e_3$, and therefore so does their sum. Since the left-hand side also satisfies the same recurrence, it suffices to check agreement at $j = 0, 1, 2$.

$j = 0$: $A\,K(n) + e_3^n \cdot 0 = A\,K(n) = K(3n)$. ✓

$j = 1$: $A\,K(n+1) + e_3^n \cdot 0 = A\,K(n+1) = K(3n+1)$, by Theorem 9.2. ✓

$j = 2$: we need $K(3n+2) = A\,K(n+2) + e_3^n$. Apply the $s_n$ formula (Lemma 5.1) at index $3n$:

$$s_{3n} = e_1\,K(3n+1) - 2e_2\,K(3n) + 3e_3\,K(3n-1).$$

Substituting $K(3n) = A\,K(n)$, $K(3n+1) = A\,K(n+1)$, and using $s_n = 0$ to replace $e_1\,K(n+1) - 2e_2\,K(n) = -3e_3\,K(n-1)$:

$$s_{3n} = 3e_3\,\big(K(3n-1) - A\,K(n-1)\big).$$

Since $x^n + y^n + z^n = 0$, Newton's identity applied to $(x^n, y^n, z^n)$ gives $s_{3n} = 3(xyz)^n = 3e_3^n$, so $K(3n-1) = A\,K(n-1) + e_3^{n-1}$. Now the recurrence at $3n+2$:

$$\begin{aligned} K(3n+2) &= e_1 \cdot A\,K(n+1) - e_2 \cdot A\,K(n) + e_3\,(A\,K(n-1) + e_3^{n-1}) \\ &= A\,\big(e_1\,K(n+1) - e_2\,K(n) + e_3\,K(n-1)\big) + e_3^n = A\,K(n+2) + e_3^n. \quad ✓ \end{aligned}$$

Uniqueness of recurrence solutions gives equality for all $j$. □

**Remark 9.4.** The values $K(0) = K(1) = 0$ are what make $j = 0$ and $j = 1$ trivially correct: the correction term $e_3^n\, K(j)$ simply vanishes there, so no computation is needed. This has nothing to do with $s_n = 0$; it is a permanent feature of $K$ inherited from the antisymmetric weights $(y - z, z - x, x - y)$. The assumption $s_n = 0$ contributes exactly one thing: it makes the ratio $K(3n)/K(n)$ equal to $K(3n + 1)/K(n + 1)$, promoting a per-index quantity into the single global constant $A$ that the identity requires. Without $s_n = 0$ no such $A$ exists.

**Remark 9.5.** The master identity can be read as a decomposition of the $3n$-shift:

- $A \cdot K(n + j)$: a scaled copy of the original sequence,
- $e_3^n \cdot K(j)$: a copy of $K$ scaled by $(xyz)^n$, encoding $s_{3n} = 3e_3^n \neq 0$.

The natural basis window for this shift is $\{K(3n), K(3n+1), K(3n+2)\}$ (indices $j = 0, 1, 2$), since it is the standard initial window of $K$ translated by $3n$. Within it, the correction $(0, 0, e_3^n)$ mirrors exactly the initial condition $(K(0), K(1), K(2)) = (0, 0, 1)$.

# 10 Constraints on a hypothetical FLT counter-example

*Disclaimer.* Fermat's Last Theorem was proved by Wiles in 1995, and nothing here comes close to an independent proof — nor is that the aim. The goal is to see what arithmetic constraints the framework imposes on a hypothetical primitive solution $x^n + y^n + z^n = 0$ with $x, y, z \in \mathbb{Z}$.

Throughout this section $x$, $y$, $z$ are pairwise coprime nonzero integers with $s_n = 0$ and $n \geq 3$ an odd prime. All results of Section 9 apply; in particular the ratio $A$ and the master identity are in force. The additional content here is purely about divisibility.

## Divisibility consequences of the master identity

Since $x$, $y$, $z$ are integers, every $K(m)$ is an integer. We now show $K(n) \mid K(3n)$, after which divisibility of $K(3n + 1)$ and the full master identity follow with no further calculation.

**Theorem 10.1.** *If $x$, $y$, $z$ are coprime integers with $s_n = 0$, then $K(n) \mid K(3n)$, and consequently $K(n + 1) \mid K(3n + 1)$, with the same integer quotient $A = K(3n)/K(n)$.*

*Proof.* We show $K(3n) \equiv 0 \pmod{K(n)}$. Applying the addition formula (Lemma 8.1) with $m = n$ and working modulo $K(n)$:

$$\begin{aligned} K(2n) &\equiv L(n)\, K(n + 1), \\ K(2n + 1) &\equiv L(n)\, K(n + 2) + M(n)\, K(n + 1), \end{aligned}$$

since every term containing $K(n)$ as a factor vanishes. A further application modulo $K(n)$, substituting $L(n) \equiv K(n+1)$, $M(n) \equiv e_3\, K(n-1)$, and $K(n+2) \equiv e_1\, K(n+1) + e_3\, K(n-1)$, gives

$$K(3n) \;\equiv\; K(n + 1)^2 \big(e_1\, K(n + 1) + 3e_3\, K(n - 1)\big) \pmod{K(n)}.$$

Since $s_n = 0$, Lemma 5.1 gives $e_1\,K(n+1) + 3e_3\,K(n-1) = 2e_2\,K(n) \equiv 0 \pmod{K(n)}$. Hence $K(3n) \equiv 0 \pmod{K(n)}$, so $A = K(3n)/K(n)$ is an integer.

Since $A$ is an integer, Theorem 9.2 gives $K(3n+1) = A\,K(n+1)$, so $K(n+1) \mid K(3n+1)$ with the same quotient $A$. □

Since $A$ is an integer, the master identity now takes on its full arithmetic force:

**Corollary 10.2** (Master identity over $\mathbb{Z}$)**.** *If $x$, $y$, $z$ are pairwise coprime nonzero integers with $s_n = 0$ and $n \geq 3$ an odd prime, then all of $K(m)$, $e_3^n$, and $A = K(3n)/K(n)$ are integers, and for all integers $j$,*

$$K(3n+j) \;=\; A\,K(n+j) \;+\; e_3^n\,K(j).$$

*In particular, $K(n) \mid K(3n+j) - e_3^n\,K(j)$ for all $j$.*

*Proof.* Integrality of $K(m)$ follows since $x, y, z \in \mathbb{Z}$. Integrality of $A$ is Theorem 10.1. The identity is Theorem 9.3; divisibility follows immediately. □

## The divisibility tower: $K(n) \mid K(mn)$ for all $m \geq 1$

We first record a consequence of the addition formula that requires no assumption on $s_n$.

**Lemma 10.3.** *For any distinct nonzero $x, y, z$ and any integer $n$, $K(n) \mid K(2n)$ unconditionally.*

*Proof.* Set $m = n$ in the addition formula (Lemma 8.1):

$$K(2n) \;=\; K(n)\,K(n+1) \;+\; L(n)\,K(n) \;-\; M(n)\,K(n-1).$$

Since $L(n) = K(n+1) - e_1K(n)$ and $M(n) = e_3K(n-1)$, substituting gives

$$K(2n) \;=\; K(n)\,\big[2K(n+1) - e_1K(n) - e_3K(n-1)\big].$$

The bracket is an integer whenever $x, y, z$ are integers, so $K(n) \mid K(2n)$. □

**Corollary 10.4** (Divisibility tower)**.** *Under the hypotheses of Corollary 10.2 (pairwise coprime integers, $s_n = 0$, $n \geq 3$ an odd prime),*

$$K(n) \;\mid\; K(mn) \qquad \textit{for all integers } m \geq 1.$$

*Proof.* We induct on $m$, with the inductive step consuming two previous cases.

*Base cases.* $m = 1$: trivial. $m = 2$: Lemma 10.3. $m = 3$: Theorem 10.1.

*Inductive step.* Suppose $m \geq 4$ and that $K(n) \mid K((m-2)n)$ and $K(n) \mid K((m-3)n)$. Apply the master identity (Corollary 10.2) with $j = (m-3)n$:

$$K(mn) \;=\; K\big(3n + (m-3)n\big) \;=\; A\,K\big((m-2)n\big) \;+\; e_3^n\,K\big((m-3)n\big).$$

Both terms on the right are divisible by $K(n)$ by hypothesis, so $K(n) \mid K(mn)$. □

**Remark 10.5.** The two components of the master identity play distinct roles in the induction. The term $A\,K((m-2)n)$ advances by two steps using the ratio $A = K(3n)/K(n)$, while $e_3^n\,K((m-3)n)$ is the correction inherited from $s_{3n} = 3e_3^n \neq 0$. The hypothesis $s_n = 0$ enters only through the $m = 3$ base case (Theorem 10.1); once secured, the induction propagates by pure algebra. In particular, $K(n) \mid K(2n)$ (Lemma 10.3) holds without any constraint on $s_n$ and would remain true for any triple.

Draft — *exploratory, may be isolated in future*

## Bounding $d = \gcd(K(n),\, K(n+1))$

We use the identities

$$s_n = e_1\,K(n+1) - 2e_2\,K(n) + 3e_3\,K(n-1), \tag{1}$$
$$3K(n+2) = 3e_1\,K(n+1) - 3e_2\,K(n) + 3e_3\,K(n-1). \tag{2}$$

Since $s_n = 0$ and $d \mid K(n), K(n+1)$, equation (1) gives

$$d \;\mid\; 3e_3\,K(n-1). \tag{3}$$

Substituting into (2) gives $d \mid 3K(n+2)$. So $d$ divides $3K(n)$, $3K(n+1)$, $3K(n+2)$: three consecutive multiples of 3. Applying the recurrence inductively, $d \mid 3K(m)$ for all $m \geq n$, and in particular $d \mid 3K(2n-1)$, $3K(2n)$, $3K(2n+1)$.

Applying (1) at index $2n$ gives $d \mid s_{2n}$. Since $s_n = 0$,

$$s_{2n} = s_n^2 - 2\,[(xy)^n]_{\text{cyc}} = -2(x^ny^n + y^nz^n + z^nx^n),$$

so

$$d \;\mid\; 2(x^ny^n + y^nz^n + z^nx^n). \tag{4}$$

**Parity.** Since $x$, $y$, $z$ are pairwise coprime with $s_n = 0$, exactly one is even: if all three were odd then $s_n \equiv 1+1+1 = 3 \equiv 1 \pmod 2 \neq 0$, and if two were even they would share the factor 2, contradicting coprimeness. Therefore $e_1, e_3$ are even and $e_2$ is odd. The recurrence mod 2 becomes $K(m+3) \equiv K(m+1) \pmod 2$, giving the alternating pattern $K(\text{even}) \equiv 1$, $K(\text{odd}) \equiv 0 \pmod 2$. Since $n$ is an odd prime, $K(n)$ is even and $K(n+1)$ is odd, so $2 \nmid d$.

We now analyse which odd prime powers can divide $d$. Let $p$ be any odd prime dividing $d$, and let $a = p^k$ be the exact power of $p$ in $d$.

**Case 1: $p \mid e_3 = xyz$.** Since $x$, $y$, $z$ are pairwise coprime, $p$ divides exactly one of them, say $p \mid x$, so $a \mid x^n$. From (4), $a \mid 2(x^ny^n + y^nz^n + z^nx^n)$. Since $a \mid x^n$ the first and third cross terms vanish mod $a$, leaving $a \mid 2y^nz^n$. Since $p$ is odd, $\gcd(a, 2) = 1$, and $\gcd(x, y) = \gcd(x, z) = 1$ gives $\gcd(a, y^nz^n) = 1$, so $a \mid 1$: contradiction. No odd prime factor of $d$ divides $e_3$.

**Case 2:** $p \nmid e_3$**.** From (3), $a \mid 3e_3K(n-1)$. Since $p \nmid e_3$, we get $a \mid 3K(n-1)$. Together with $a \mid 3K(n)$ and $a \mid 3K(n+1)$, backward induction on the recurrence

$$3K(m) = 3e_1\,K(m+1) - 3e_2\,K(m+2) + 3e_3\,K(m+3)$$

(valid at each step since $p \nmid e_3$) gives $a \mid 3K(m)$ for all $m \leq n$. At $m = 2$: $a \mid 3K(2) = 3$, so $p^k \mid 3$, hence $p = 3$ and $k = 1$.

Combining: $d$ is odd (parity), Case 1 is vacuous for odd primes, and Case 2 forces every prime power in $d$ to divide 3.

**Proposition 10.6.** *If $x$, $y$, $z$ are pairwise coprime nonzero integers with $s_n = 0$ and $n$ an odd prime, then* $\gcd(K(n),\, K(n+1))$ *divides* 3.

**Remark 10.7.** Working with the exact prime power $a = p^k$ handles higher powers such as 9 in the same stroke: $p^k \mid 3$ forces $k = 1$ immediately, with no separate argument needed.